

Semiconductor Device Modelling

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Preliminaries

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Axioms – Quantum Mechanics

- In Quantum Mechanics the variables that describe the system are represented by a mathematical operators

Variable	Mathematical Operator
Position, x	$\hat{x} \equiv x$
Momentum, p_x	$\hat{p}_x \equiv \frac{\hbar}{i} \frac{\partial}{\partial x}$
Total Energy, E	$\hat{E} \equiv -\frac{\hbar}{i} \frac{\partial}{\partial t}$
Potential Energy, $U(x)$	$\hat{U} \equiv U(x)$

- There exist a function/variable $\Psi(x, t)$ from which all the observable quantities can be extracted and it is called the wavefunction.

Axioms – Quantum Mechanics

- The total energy of a particle

$$\frac{p^2}{2m} + U = E$$

- Using the operators on the wavefunction we get

$$-\frac{\hbar^2}{2m} \nabla^2 \Psi(\mathbf{r}, t) + U(\mathbf{r}) \Psi(\mathbf{r}, t) = -\frac{\hbar}{i} \frac{\partial}{\partial t} \Psi(\mathbf{r}, t)$$

- If we assume that the system is in steady state: $\Psi(\mathbf{r}, t) = \psi(\mathbf{r})\phi(t)$
- Using the above separation of variables we end up with the time independent form of Schrödinger equation

$$-\frac{\hbar^2}{2m} \nabla^2 \psi(\mathbf{r}) + U(\mathbf{r}) \psi(\mathbf{r}) = E \psi(\mathbf{r})$$

Axioms – Quantum Mechanics

- $\psi(\mathbf{r}, t)$ and $\nabla\psi(\mathbf{r}, t)$ are continuous (Infinite well problem).
- Probability of locating an electron in a small volume is $|\psi(\mathbf{r}, t)|^2 d\mathbf{r}$
- Normalization condition

$$\int |\psi(\mathbf{r}, t)|^2 d\mathbf{r} = 1$$

- Since we are dealing in terms of probabilities, the expectation value of any variable, ζ is given by

$$\langle \zeta \rangle = \int \psi^\dagger(\mathbf{r}) \zeta \psi(\mathbf{r}) d\mathbf{r}$$

Schrödinger's Equation - Infinite Well

- The potential is assumed to be infinite for $x < 0$ and $x > L_x$

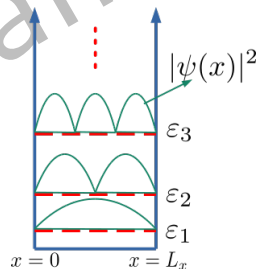
$$\frac{-\hbar^2}{2m} \nabla^2 \psi(x, t) + U(x, t) \psi(x, t) = E \psi(x, t)$$

- $U(x) = 0$ in the well and infinite outside
- General solution

$$\psi(x, t) = A \sin(kx) + B \cos(kx)$$

- Boundary conditions:
 $\psi(x, t) = 0$ @ $x = 0$ & $x = L_x$
- Applying boundary condition to the general solution gives

$$\psi(x, t) = A \sin(kx); \quad k_n = \frac{n\pi}{L_x} \quad n = 1, 2, 3...$$



Schrödinger's Equation - Infinite Well

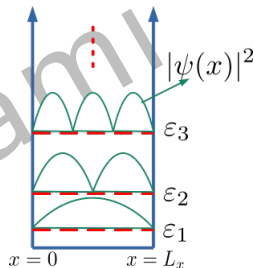
- Normalization condition: $\int |\psi(x)|^2 dx = 1$

$$A = \sqrt{\frac{2}{L_x}}$$

- Energy in terms of k

$$E_n = \frac{\hbar^2}{2m} k_n^2 = \frac{\hbar^2}{2m} \frac{n^2 \pi^2}{L_x^2}$$

- Discontinuity of $\frac{\partial \psi(x)}{\partial x}$ at boundaries ?
- Lowest energy, E_1 is greater than 0.



$$U = 0 \quad 0 < x < L_x$$

$$U = \infty \quad x \leq 0 \text{ or } x \geq L_x$$

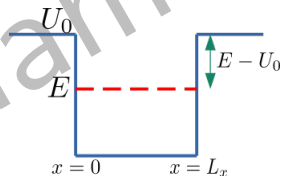
Schrödinger's Equation - Finite Well

- The potential is finite bounded between $(0, L_x)$
- Schrödinger's equation in the well

$$\frac{-\hbar^2}{2m} \nabla^2 \psi(x) = E \psi(x)$$

$$\nabla^2 \psi(x) = k^2 \psi(x) \quad k = \sqrt{\frac{2mE}{\hbar^2}}$$

$$\psi_W(x) = A_W \sin(kx) + B_W \cos(kx)$$



- Outside the well

$$\nabla^2 \psi(x) = \kappa^2 \psi(x) \quad \kappa = \sqrt{\frac{2m(U_0 - E)}{\hbar^2}}$$

$$U = 0 \quad 0 < x < L_x$$

$$U = U_0 \quad x \leq 0 \quad x \geq L_x$$

$$\psi_{<}(x) = A_{<} e^{\kappa x} + B_{<} e^{-\kappa x}$$

$$\psi_{>}(x) = A_{>} e^{\kappa x} + B_{>} e^{-\kappa x}$$

Schrödinger's Equation - Finite Well

- Boundary conditions

$$\psi(-\infty) = \psi(\infty) = 0$$

- This results in

$$\psi_{<}(x) = A_{<}e^{\kappa x} \quad \psi_{>}(x) = B_{>}e^{-\kappa x}$$

- The continuity of the wavefunction demands

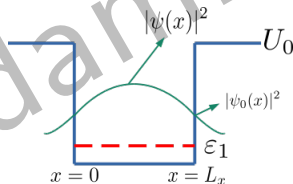
$$\psi_{<}(0) = \psi_W(0) \quad \psi_W(L_x) = \psi_{>}(L_x)$$

$$\frac{\partial}{\partial x}\psi_{<}(0) = \frac{\partial}{\partial x}\psi_W(0) \quad \frac{\partial}{\partial x}\psi_W < (L_x) = \frac{\partial}{\partial x}\psi_{>}(L_x)$$

- This results in

$$A_{<} = B_W; \quad A_W \sin(ka) + B_W \cos(ka) = B_{>}e^{-\kappa L_x}$$

$$\kappa A_{<} = kA_W; \quad kA_W \cos(ka) - kB_W \sin(ka) = -\kappa B_{>}e^{-\kappa L_x}$$



Schrödinger's Equation - Finite Well

- Solving the above system of equations we get

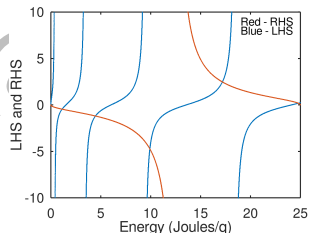
$$A_W [(k^2 - \kappa^2) \sin(kL_x) - 2\kappa k \cos(ka)] = 0$$

- For a non trivial solutions

$$(k^2 - \kappa^2) \sin(kL_x) - 2\kappa k \cos(ka) = 0$$

$$\tan(kL_x) = 2 \frac{\kappa k}{k^2 - \kappa^2}$$

- Solution of the above equation:
 - Graphical solution
 - Numerical solution – Newton-Raphson method



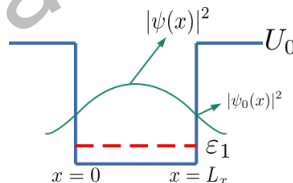
$$L_x = 0.5 \text{ nm}$$

$$U_0 = 25q \text{ J} = 25\text{eV}$$

Schrödinger's Equation - Finite Well

General Features of the solution

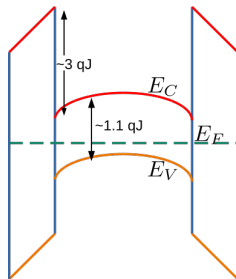
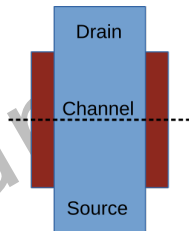
- In the limit $U_0 \rightarrow \infty$ then the energy values tend to the infinite potential well.
- The $\psi(x)$ is sinusoidal in the well.
- The wavefunction has an exponential tail outside the well
- As $|\psi(x)|^2 dx$ denotes the probability then it is possible to find the particle in the classically forbidden well (Tunneling)



Numerical solution of the Schrödinger's equation.

Practical Applications

- The expected value of the electron distribution lies slightly (≈ 1 nm) away from the oxide semiconductor interface thus reducing the effective gate control (gate capacitance)
- The semiconductor region plays the role of the potential well
- The oxide-semiconductor barrier have finite value but may be also be modelled as infinite barrier by making $\psi(x) = 0$ at the interface
- If the oxide barrier is thin as compared to the wavefunction spread in the barrier region then an electron may leak out giving rise to the gate leakage current.



Electronic Energy levels

- In the early 20 century it was observed the the atomic spectra was well defined discrete levels.
- This was amazingly explained by Neils Bohr by assuming that the electrons can only take certain orbits. (Note He didn't had the luxury of the using Schrödinger's equation. It was intuition and pure genius)
- But now we know that the use of Schrödinger's equation gives rise to discrete energy levels.

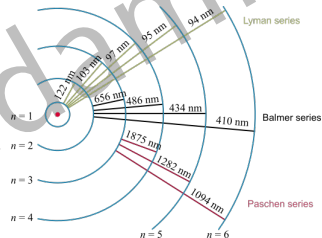


Figure: Hydrogen Spectra.
Ref: Wikipedia

Electronic Energy Levels

- When two atoms are brought in close proximity the atoms start to interact (overlapping of wavefunctions) and this leads to splitting of the energy levels – Pauli's Exclusion Principle.
 - Bonding energy level
 - Anti-bonding energy level
- As the number of atoms that are brought in close proximity increases the energy levels get closer and closer leading to formation of bands
 - Valence band
 - Conduction band
- A *band* is essentially a set of discrete energy levels that are so close to each other that their spacing doesn't matter

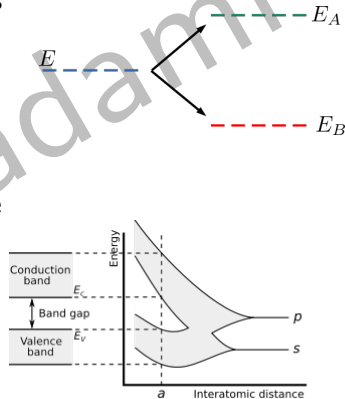
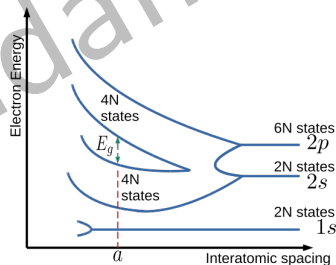


Figure: Ref: Wikipedia

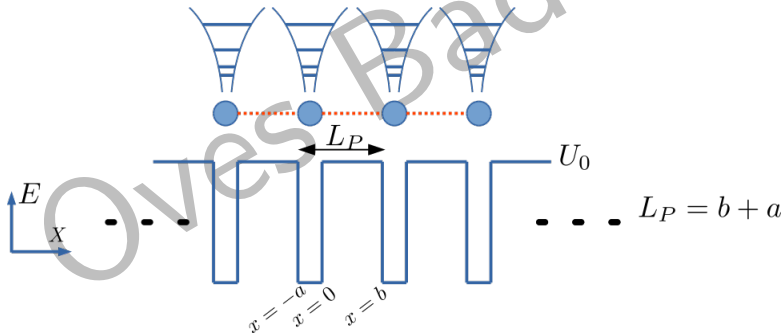
A Specific Case of Diamond

- Diamond atom has 4 electrons in its highest energy level ($2s2p$)
- When N atoms are brought close to each other $2s$ and $6p$ (p_x , p_y and p_z) energy levels combine to form a new set of states having 2 pairs of 4 states.
- This process is called a " sp^3 hybridisation".
- An equilibrium is attained when the energy of the system (crystal) is minimized.
- The $4N$ electrons occupy the lower energy states (called *Valence Band*)
- Higher $4N$ states remain empty (*Conduction Band*)



Kronig-Penney Model – Electronic Bandstructure

- A repetitive 1D arrangement of atoms.
- A potential well of width a and barrier of width b
- Continuity of the wavefunction at the interfaces
- What about the periodicity of the structure – Can it be used ?



A Digression to Bloch's Theorem

- In a periodic structure the wavefunction takes a form

$$\psi_k(x) = e^{ik \cdot x} u_k(x)$$

$$u_k(r + L_p) = u_k(x)$$

- Putting these together

$$\psi_k(x + L_P) = e^{ik \cdot (x + L_P)} u_k(x + L_P) = e^{ik \cdot L_P} e^{ik \cdot x} u_k(x) = e^{ik \cdot L_P} \psi_k(x)$$

- Note that $\psi(x)$ is not periodic as it might seem logical.
- To determine k one needs to apply periodic BC (also called as Born-von Karman BC) – N_A is the total number of atoms in the 1D lattice

$$\psi(r) = \psi(r + N_A L_P) = e^{iN_A k L_P} \psi(r)$$

- To satisfy this: $e^{iN_A k L_P} = 1$

$$k = n \frac{2\pi}{N_A L_P} \quad n = 0, \pm 1, \pm 2, \dots$$

Bloch's Theorem – A phenomenological proof

- Any function can be written as a product of 2 functions

$$\psi(x) = f(x)u(x)$$

where $f(x)$ is some unknown function and $u(x)$ is a periodic function with a period of L_p

- We first of all note that the probability of finding an electron at a point x and $x + L_p$ should be the same.
- The probability distribution of electron is $|\psi(x)|^2$. Thus

$$|\psi(x)|^2 = |\psi(x + L_p)|^2$$

$$|f(x)| = |f(x + L_p)|$$

- A function that satisfies this requirement is an exponential function of the form e^{ikx}

Kronig-Penney Model – Electronic Bandstructure

- Wavefunctions are of the form

$$\psi_W(x) = A_W \sin(k_W x) + B_W \cos(k_W x)$$

$$\psi_B(x) = A_B \sin(k_B x) + B_B \cos(k_B x)$$

$$k_W = \sqrt{\frac{2mE}{\hbar^2}}; \quad k_B = \sqrt{\frac{2m(E - U_0)}{\hbar^2}}$$

- Continuity conditions at $x = 0$

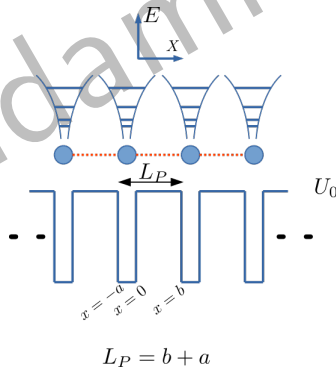
$$\psi_W(0) = \psi_B(0)$$

$$\left. \frac{d\psi_W}{dx} \right|_0 = \left. \frac{d\psi_B}{dx} \right|_0$$

- Periodicity condition (Bloch's theorem) dictates

$$\psi_W(-a) = e^{-ik(b+a)} \psi_B(b)$$

$$\left. \frac{d\psi_W}{dx} \right|_{-a} = e^{-ik(b+a)} \left. \frac{d\psi_B}{dx} \right|_b$$



Kronig-Penney Model – Electronic Bandstructure

- Solving the above equations we get the following relationship

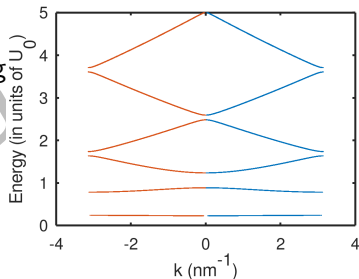
$$-\frac{k_W^2 + k_B^2}{2k_W k_B} \sin(k_W a) \sin(k_B b)$$

$$+ \cos(k_W a) \cos(k_B b) = \cos(kL_p)$$

- The above equation can be solved using

- Graphical method
- Numerical method

- Allowed energy levels are called as bands and excluded energy levels are forbidden energy gaps.
- Energy bands naturally arise in a periodic atomic arrangements
- Electronic bandstructure is restricted to $k = \pm \frac{\pi}{L_p}$ – First Brillouin zone

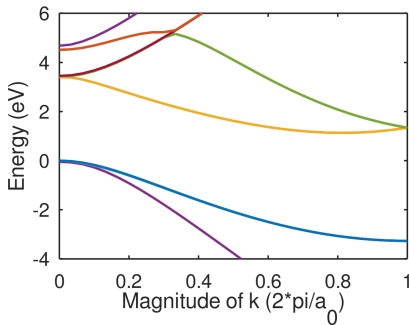


$$a = b = 0.5 \text{ nm}$$

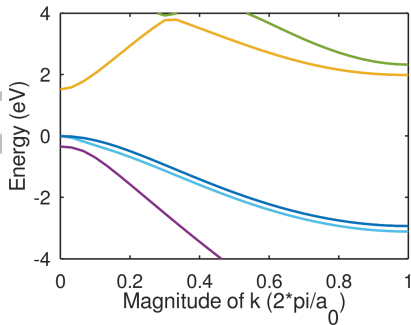
$$U_0 = 3 \text{ qJ} = 3 \text{ eV}$$

Electronic Bandstructures for Si and GaAs

- Silicon



- GaAs



- The allowed energy levels are like chairs in a banquet hall

Electronic Bandstructure - Effective Mass

- Bloch theorem indicates that the wavefunctions($u_k(x)e^{ikx}$) are of the form of waves.

$$\nu = \frac{d\omega}{dk} = \frac{1}{\hbar} \frac{dE}{dk} = \frac{dE}{dp_c}$$

where $p_c = \hbar k$ is the *crystal momentum*

- Consider an electron moving an electric field $\mathcal{E}$, then rate at which the electron will gain energy is

$$\frac{dE}{dt} = -q\mathcal{E}\nu = \frac{F}{\hbar} \frac{dE}{dk} = \frac{F}{\hbar} \frac{dE}{dt} \frac{dt}{dk} \rightarrow F = \frac{dp_c}{dt}$$

$$a = \frac{d\nu}{dt} = \frac{1}{\hbar} \frac{d}{dt} \left(\frac{dE}{dk} \right) = \frac{1}{\hbar^2} \frac{d^2E}{dk^2} \frac{d(\hbar k)}{dt} = \frac{1}{\hbar^2} \frac{d^2E}{dk^2} F$$

$$F = m^* \frac{d\nu}{dt} \quad m^* = \frac{\hbar^2}{d^2E/dk^2} \quad \text{Newton's second law}$$

Effective mass: A Physical Picture

- The actual momentum is given by

$$p = m_0 v$$

where m_0 is the free electron mass

- Thus the net force, F_{net} , acting on the electrons is

$$F_{net} = m_0 \frac{dv}{dt} = F_{ext} + F_L$$

F_{ext} – external force due to applied bias

F_L – force due to the crystal lattice

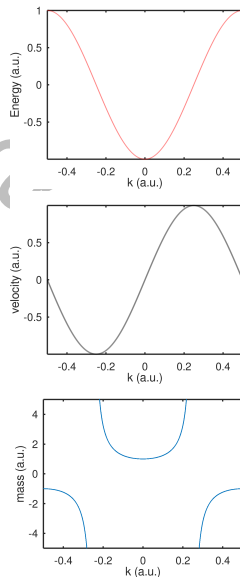
$$m_0 \frac{dv}{dt} = m_0 \frac{F_{ext}}{m^*}$$

$$\Rightarrow m^* = m_0 \frac{F_{ext}}{F_{ext} + F_L}$$

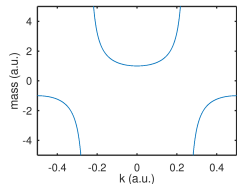
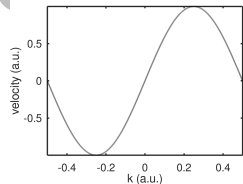
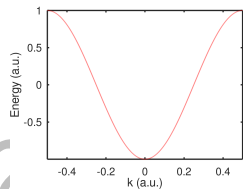
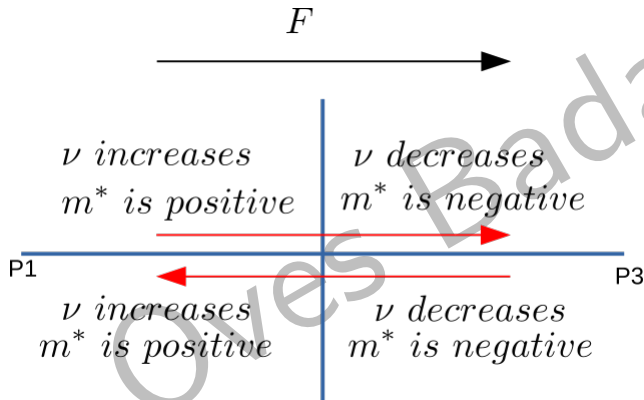
- Thus depending on the configuration (band structure and occupation of the electrons) the m^* may be greater or smaller than m_0 .

Electronic Bandstructure - Bloch Oscillations

- Consider a single electron in an empty band at $k = 0$ when we apply an electric field.
- The electron starts to gain energy and moves higher in the band.
- After reaching the maximum velocity starts to decelerate.
- The effective mass of the electron becomes negative after the velocity peaks.
- When the electron reaches the right end then its velocity becomes negative and it appears at the left edge.
- Thus an electron continues to oscillate and hence the net current is zero. These oscillations are called as *Bloch oscillations*.

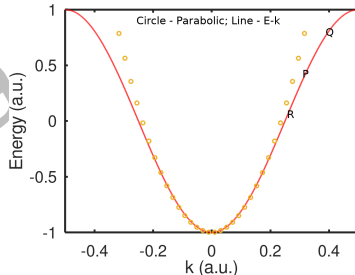


Bloch Oscillations



Electronic Bandstructure - Impact of Scattering

- Consider an electron at the energy position P in the electric field.
- The electron then can reach the state Q but most of the time the electron collides with lattice (or any mechanism that breaks the lattice periodicity) and loses the momentum and goes to the state R .
- Generally the electron stays near the bottom of the band due to scattering.
- At the minima/maxima the band can be approximated with a parabola (i.e. $E = m^*k^2$). This parabolic band approximation.



Density of States (DOS)

- Carriers in a crystal can occupy only certain energy levels.
- DOS is defined as the number of available energy states in a crystal per unit volume per unit energy (units $m^{-3}J^{-1}$). Please note that the DOS is defined for a particular band.
- Let us assume that the DoS effective mass of the band is given by m^* (we are assuming that the parabolic effective mass approximation i.e. the effective mass is constant for the entire energy band).
- Let the dimensions of the crystal having dimensions of L_x , L_y and L_z .
- Schrodinger's equation (more correctly the effective mass equation) for this system can be written as (without the loss of generality we can set the potential energy to 0)

$$-\frac{\hbar^2}{2m^*}\nabla^2\psi(x,y,z) = E\psi(x,y,z) \rightarrow \nabla^2\psi(x,y,z) + k^2\psi(x,y,z) = 0$$

- An electron in a crystal behaves like a normal electron except for its mass.

Density of States (DOS)

- To solve the above equation we assume that the wavefunction can be written in the form $\psi(x, y, z) = \psi(x)\psi(y)\psi(z)$.
- Substituting we get

$$\frac{1}{\psi(x)} \frac{\partial^2 \psi(x)}{\partial x^2} + \frac{1}{\psi(y)} \frac{\partial^2 \psi(y)}{\partial y^2} + \frac{1}{\psi(z)} \frac{\partial^2 \psi(z)}{\partial z^2} = -k^2$$

- Since the RHS is a constant and each of the LHS terms depend only on a single parameters, the above equation will be satisfied only when

$$\frac{1}{\psi(x)} \frac{\partial^2 \psi(x)}{\partial x^2} = -k_x^2 = \text{constant}$$

Similar equations can also be written for terms along y and z directions

Density of States (DOS)

- Thus the wavefunction along x is

$$\psi(x) = A_x \sin(k_x x);$$

- In practice, the crystals are very large (compared to the unit cell) and hence we can assume that the wavefunction is periodic

$$\psi(x) = \psi(x + L_x) \implies k_x = \frac{2\pi n_x}{L_x}$$

- Thus the complete wavefunction and wavevector can be written as

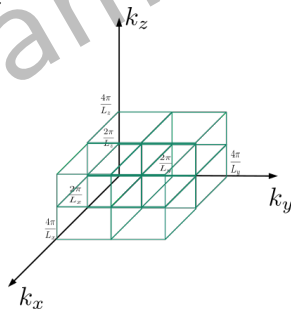
$$\psi(x, y, z) = A \sin(k_x x) \sin(k_y y) \sin(k_z z); \quad k^2 = k_x^2 + k_y^2 + k_z^2$$

where A is the constant that can be determined by the normalization of the wavefunction

Density of States

- Each combination of $\mathbf{k} = (k_x, k_y, k_z)$ represents a state that one electron can occupy (we will account for the spin degeneracy later).
- The volume of the each state in this k - space is $\frac{(2\pi)^2}{L_x L_y L_z} = \frac{(2\pi)^2}{V}$
- The volume of the spherical shell enclosed between k and $k + dk$ is given by $4\pi^2 k^2 dk$
- Number of states in the spherical shell, $N(k)dk$, is $\frac{k^2 dk V}{2\pi^2}$
- Thus the number of states $N(k)$ per unit volume is $\frac{k^2 dk}{2\pi^2}$.
- We can simplify it further by recollecting that

$$k^2 = \frac{2m^* E}{\hbar^2}; \quad \text{and} \quad dk = \sqrt{\frac{m^*}{\hbar^2}} \frac{dE}{\sqrt{2E}}$$



Density of States

- Substituting the k^2 and dk in the number of states per unit volume, and multiplying by 2 to account for spin degeneracy

$$N(E)dE = \frac{(m^*)^{3/2}(2E)^{1/2}}{\pi^2\hbar^3}dE$$

- Thus the number of states per unit energy per unit volume, *Density of States* ($N(E)$), can be written as

$$N(E) = \frac{(m^*)^{3/2}(2E)^{1/2}}{\pi^2\hbar^3}$$

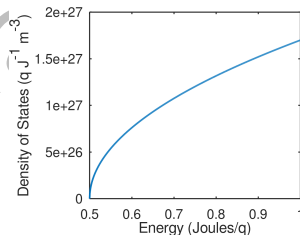
- Depending upon if the density of states is being calculated for conduction or valence band the appropriate effective masses must be employed.

Density of States

- Accounting for the minimum(maximum) possible energy level, $E_c(E_v)$, that can be occupied for conduction(valence) band, the density of states can be calculated as

$$N(E) = \frac{(m_n^*)^{3/2} (2(E - E_c))^{1/2}}{\pi^2 \hbar^3} \quad E \geq E_c$$

$$N(E) = \frac{(m_v^*)^{3/2} (2(E_v - E))^{1/2}}{\pi^2 \hbar^3} \quad E \leq E_v$$



$$E_c = 0.5q J$$

- The 3D density of states increases monotonically with energy.
- The above derivation is for 3D structures. Can you extend it for 2D and 1D structure?

Occupation Statistics

- Electrons in a semiconductor crystal are indistinguishable particles and they obey Pauli's exclusion principle.
- The probability that an electron will occupy a particular energy level is given by Fermi-Dirac distribution function.

$$f(E) = \frac{1}{1 + e^{(E-F_F)/(k_B T)}}$$

E_F is the Fermi Energy.

- In the approximation that $E \gg E_F$ the Fermi-Dirac distribution reduces to Maxwell-Boltzmann distribution

$$f(E) = e^{-(E-F_F)/(k_B T)}$$

